

FLIGHT DELAY ANALYSIS

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project overview:

This project aims to analyze flight delay data to understand the main factors affecting delays.

The goal is to identify patterns and insights from real flight data.





PROJECT MANAGEMENT TIMELINE

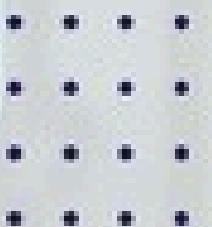
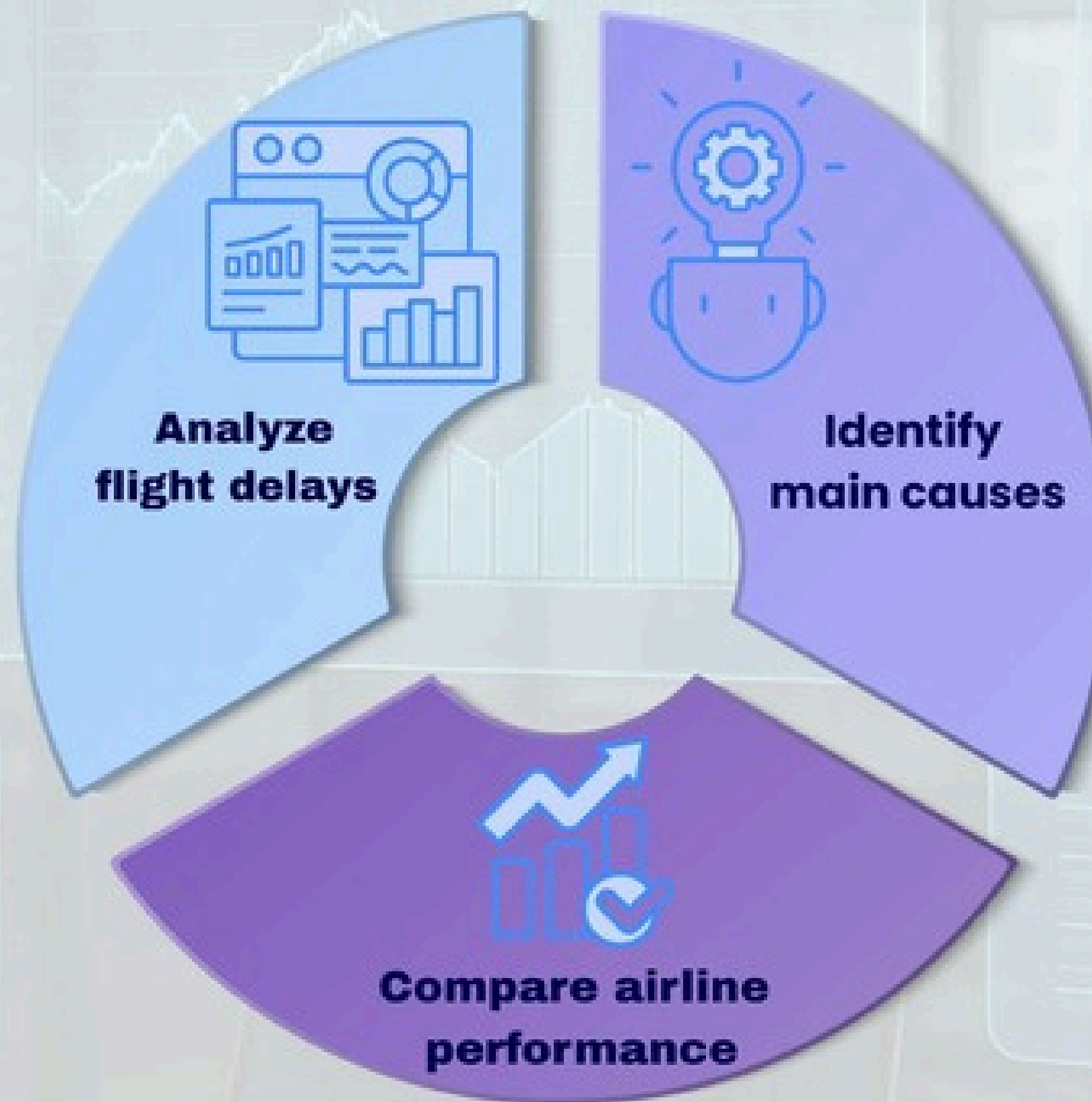
Flight Delay Analysis Project

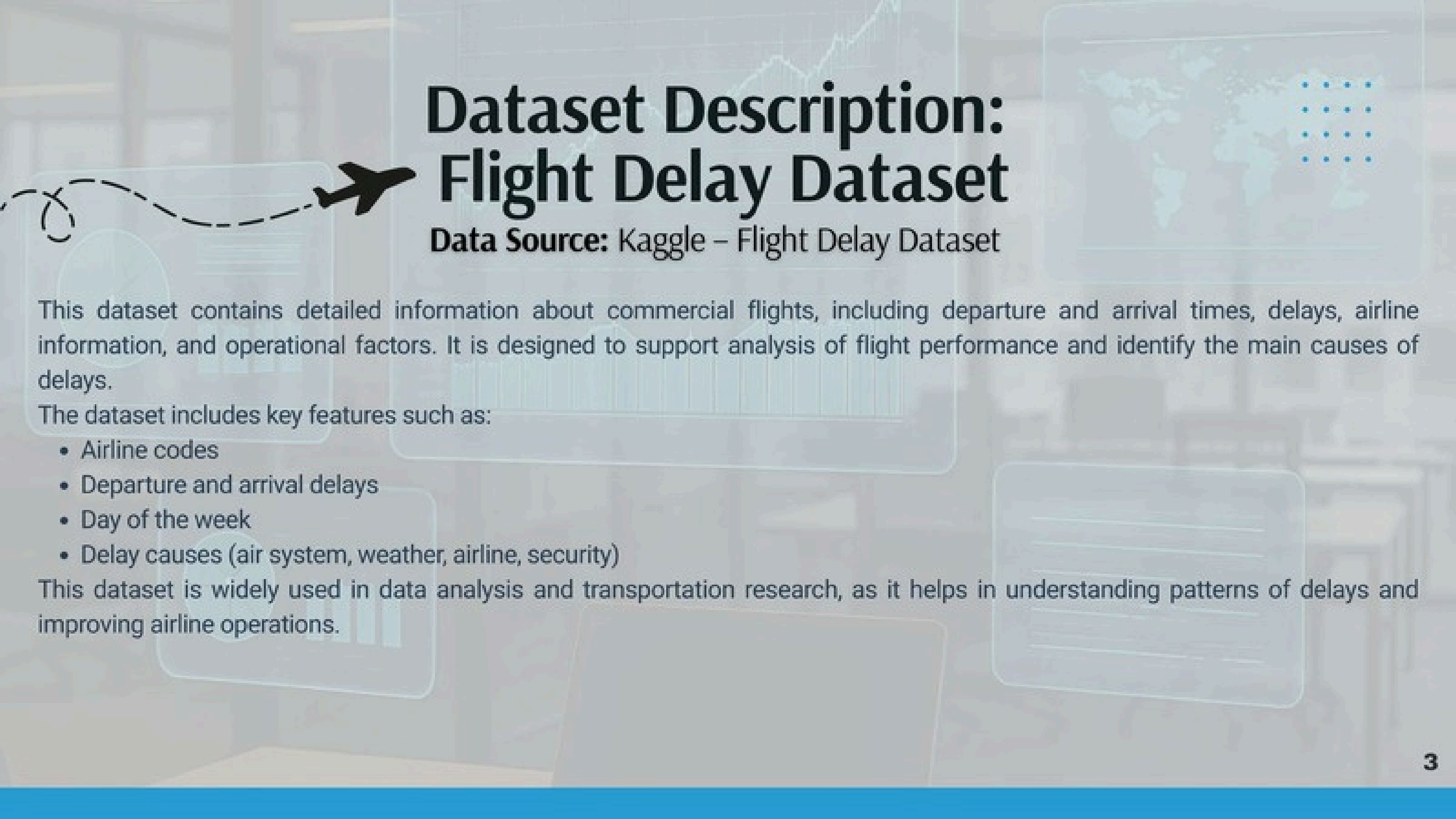
PHASE	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5	DAY 6
 Data Cleaning Clean and prepare the dataset	1 - 2 Days					
 Data Analysis Analyze data and find patterns & trends		2 Days				
 Visualization Create charts and visualizations				1 Day		
 Reporting Write report and extract key insights					1 Day	



🕒 Total Duration: 6 Days

Objective:





Dataset Description: Flight Delay Dataset

Data Source: Kaggle – Flight Delay Dataset

This dataset contains detailed information about commercial flights, including departure and arrival times, delays, airline information, and operational factors. It is designed to support analysis of flight performance and identify the main causes of delays.

The dataset includes key features such as:

- Airline codes
- Departure and arrival delays
- Day of the week
- Delay causes (air system, weather, airline, security)

This dataset is widely used in data analysis and transportation research, as it helps in understanding patterns of delays and improving airline operations.

Reading the dataset:

Data Source: Kaggle – Flight Delay Dataset

```
In [1]: pip install Kagglehub
```

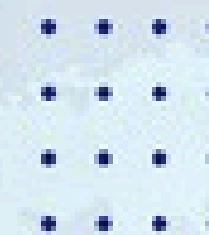
```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: Kagglehub in c:\users\dell\appdata\roaming\python\python39\site-packages (0.3.13)
Requirement already satisfied: requests in c:\programdata\anaconda3\lib\site-packages (from Kagglehub) (2.28.1)
Requirement already satisfied: tqdm in c:\programdata\anaconda3\lib\site-packages (from Kagglehub) (4.64.1)
Requirement already satisfied: packaging in c:\programdata\anaconda3\lib\site-packages (from Kagglehub) (21.3)
Requirement already satisfied: pyyaml in c:\programdata\anaconda3\lib\site-packages (from Kagglehub) (6.0)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in c:\programdata\anaconda3\lib\site-packages (from packaging->Kagglehub) (3.0.9)
Requirement already satisfied: idna<4,>=2.5 in c:\programdata\anaconda3\lib\site-packages (from requests->Kagglehub) (3.3)
Requirement already satisfied: urllib3<1.27,>=1.21.1 in c:\programdata\anaconda3\lib\site-packages (from requests->Kagglehub) (1.26.11)
Requirement already satisfied: certifi>=2017.4.17 in c:\programdata\anaconda3\lib\site-packages (from requests->Kagglehub) (2022.9.14)
Requirement already satisfied: charset-normalizer<3,>=2 in c:\programdata\anaconda3\lib\site-packages (from requests->Kagglehub) (2.0.4)
Requirement already satisfied: colorama in c:\programdata\anaconda3\lib\site-packages (from tqdm->Kagglehub) (0.4.5)
Note: you may need to restart the kernel to use updated packages.
```

```
In [2]: import pandas as pd
import numpy as np
```

```
In [29]: df=pd.read_csv("C:\\Users\\DELL\\Desktop\\Jouhara\\flights.csv")
```



INITIAL DATA EXPLORATION TO DISPLAY FIRST 5 ROWS OF THE DATASET USE DF.HEAD()



```
df.head()
```

	YEAR	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE	...
0	2015	1	1	4	AS	98	N407AS	ANC	SEA	5	...
1	2015	1	1	4	AA	2336	N3KUAA	LAX	PBI	10	...
2	2015	1	1	4	US	840	N171US	SFO	CLT	20	...
3	2015	1	1	4	AA	258	N3HYAA	LAX	MIA	20	...
4	2015	1	1	4	AS	135	N527AS	SEA	ANC	25	...

5 rows x 31 columns

4

Data Cleaning Process

```
columns_needed = [  
    'AIRLINE',  
    'DEPARTURE_DELAY',  
    'ARRIVAL_DELAY',  
    'DAY_OF_WEEK',  
    'SCHEDULED_DEPARTURE',  
    'DISTANCE'  
]
```

```
df_small = df[columns_needed]
```

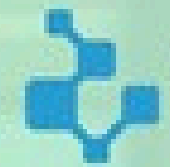
df_small

	AIRLINE	DEPARTURE_DELAY	ARRIVAL_DELAY	DAY_OF_WEEK	SCHEDULED_DEPARTURE	DISTANCE
0	AS	-11.0	-22.0	4	5	1448
1	AA	-8.0	-9.0	4	10	2330
2	US	-2.0	5.0	4	20	2296
3	AA	-5.0	-9.0	4	20	2342
4	AS	-1.0	-21.0	4	25	1448
...	...	...	...	...	...	...
5819074	B6	-4.0	-26.0	4	2359	2611
5819075	B6	-4.0	-16.0	4	2359	1617
5819076	B6	-9.0	-8.0	4	2359	1598
5819077	B6	-6.0	-10.0	4	2359	1189
5819078	B6	15.0	2.0	4	2359	1576

Data Cleaning Process

```
df_small.columns
```

```
Index(['AIRLINE', 'DEPARTURE_DELAY', 'ARRIVAL_DELAY', 'DAY_OF_WEEK',  
      'SCHEDULED_DEPARTURE', 'DISTANCE'],  
      dtype='object')
```



Show dataset information & descriptive statistics

```
df_small.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 5819879 entries, 0 to 5819878  
Data columns (total 6 columns):  
#   Column                Dtype  
---  ---  
0   AIRLINE                object  
1   DEPARTURE_DELAY        float64  
2   ARRIVAL_DELAY          float64  
3   DAY_OF_WEEK            int64  
4   SCHEDULED_DEPARTURE    int64  
5   DISTANCE               int64  
dtypes: float64(2), int64(3), object(1)  
memory usage: 266.4+ MB
```

```
df_small.describe()
```

	DEPARTURE_DELAY	ARRIVAL_DELAY	DAY_OF_WEEK	SCHEDULED_DEPARTURE	DISTANCE
count	5.732926e+06	5.714008e+06	5.819079e+06	5.819079e+06	5.819079e+06
mean	9.370158e+00	4.407057e+00	3.926941e+00	1.329602e+03	8.223565e+02
std	3.708094e+01	3.927130e+01	1.988845e+00	4.837518e+02	6.077843e+02
min	-8.200000e+01	-8.700000e+01	1.000000e+00	1.000000e+00	2.100000e+01
25%	-5.000000e+00	-1.300000e+01	2.000000e+00	9.170000e+02	3.730000e+02
50%	-2.000000e+00	-5.000000e+00	4.000000e+00	1.325000e+03	6.470000e+02
75%	7.000000e+00	8.000000e+00	6.000000e+00	1.730000e+03	1.062000e+03
max	1.988000e+03	1.971000e+03	7.000000e+00	2.359000e+03	4.983000e+03

Check if there are null values then fill the missing values with mean

```
df_small.isnull().sum()
```

AIRLINE	0
DEPARTURE_DELAY	86153
ARRIVAL_DELAY	105071
DAY_OF_WEEK	0
SCHEDULED_DEPARTURE	0
DISTANCE	0

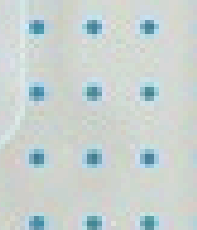
dtype: int64

```
df_small.isnull().sum().sort_values()
```

AIRLINE	0
DAY_OF_WEEK	0
SCHEDULED_DEPARTURE	0
DISTANCE	0
DEPARTURE_DELAY	86153
ARRIVAL_DELAY	105071

dtype: int64

```
df_small['ARRIVAL_DELAY'] = df_small['ARRIVAL_DELAY'].fillna(df_small['ARRIVAL_DELAY'].mean())  
df_small['DEPARTURE_DELAY'] = df_small['DEPARTURE_DELAY'].fillna(df_small['DEPARTURE_DELAY'].mean())
```



TO COUNT UNIQUE VALUES IN CATEGORICAL COLUMNS

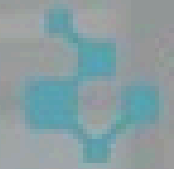
```
df_small['AIRLINE'].unique()
```

```
array(['AS', 'AA', 'US', 'DL', 'NK', 'UA', 'HA', 'B6', 'OO', 'EV', 'MQ',  
      'F9', 'WN', 'VX'], dtype=object)
```

TO COUNT DELY TIME PER DAY WEEK & AIRLINE:

```
print(df_small['AIRLINE'].value_counts())  
  
print(df_small['DAY_OF_WEEK'].value_counts())
```

```
WN      1261855  
DL       875881  
AA       725984  
OO       588353  
EV       571977  
UA       515723  
MQ       294632  
B6       267048  
US       198715  
AS       172521  
NK       117379  
F9        90836  
HA        76272  
VX        61903  
Name: AIRLINE, dtype: int64  
4        872521  
1        865543  
5        862209  
3        855897  
2        844600  
7        817764  
6        700545  
Name: DAY_OF_WEEK, dtype: int64
```



TO COUNT MEAN FOR EACH AIRLINE & DEPARTMENT DELAY:



```
print("ARRIVAL_DELAY mean",df_small['ARRIVAL_DELAY'].mean())  
print("DEPARTURE_DELAY mean",df_small['DEPARTURE_DELAY'].mean())  
print("ARRIVAL_DELAY mean",df_small.groupby('AIRLINE')['ARRIVAL_DELAY'].mean())
```

```
ARRIVAL_DELAY mean 4.407057357998102  
DEPARTURE_DELAY mean 9.37015827520495  
ARRIVAL_DELAY mean AIRLINE  
AA      3.468550  
AS     -0.942799  
B6      6.635293  
DL      0.213765  
EV      6.519779  
F9     12.438204  
HA      2.030313  
MQ      6.347611  
NK     14.284359  
OO      5.817438  
UA      5.415779  
US      3.722052  
VX      4.734207  
WN      4.375458  
Name: ARRIVAL_DELAY, dtype: float64
```

DELAY TIME ANALYSIS:

```
df_small.loc[df_small['ARRIVAL_DELAY'] > 0]
```

	AIRLINE	DEPARTURE_DELAY	ARRIVAL_DELAY	DAY_OF_WEEK	SCHEDULED_DEPARTURE	DISTANCE	DELAY_STATUS
2	US	-2.0	5.0	4	20	2296	Delayed
5	DL	-5.0	8.0	4	25	1589	Delayed
14	DL	-1.0	8.0	4	40	1399	Delayed
20	NK	25.0	6.0	4	55	1139	Delayed
21	AA	-8.0	1.0	4	100	2724	Delayed
...	...	...	...	...	...	...	...
5819057	NK	13.0	28.0	4	2358	982	Delayed
5819069	B6	159.0	159.0	4	2359	2248	Delayed
5819071	B6	16.0	17.0	4	2359	1931	Delayed
5819073	B6	16.0	3.0	4	2359	1826	Delayed
5819078	B6	15.0	2.0	4	2359	1576	Delayed

DELAY TIME ANALYSIS:

```
df_small.loc[df_small['ARRIVAL_DELAY'] < 0]
```

	AIRLINE	DEPARTURE_DELAY	ARRIVAL_DELAY	DAY_OF_WEEK	SCHEDULED_DEPARTURE	DISTANCE	DELAY_STATUS
0	AS	-11.0	-22.0	4	5	1448	On Time
1	AA	-8.0	-9.0	4	10	2330	On Time
3	AA	-5.0	-9.0	4	20	2342	On Time
4	AS	-1.0	-21.0	4	25	1448	On Time
6	NK	-6.0	-17.0	4	25	1299	On Time
...	...	...	...	...	...	...	...
5819072	B6	7.0	-11.0	4	2359	1626	On Time
5819074	B6	-4.0	-26.0	4	2359	2611	On Time
5819075	B6	-4.0	-16.0	4	2359	1617	On Time
5819076	B6	-9.0	-8.0	4	2359	1598	On Time
5819077	B6	-6.0	-10.0	4	2359	1189	On Time

DATA ANALYSIS: CHECKING DELAY STATUS

```
import numpy as np

df_small["DELAY_STATUS"] = np.where(df_small["ARRIVAL_DELAY"] > 0, "Delayed", "On Time")

df_small["DELAY_STATUS"].value_counts()

On Time    3627112
Delayed    2191967
Name: DELAY_STATUS, dtype: int64
```

DATA ANALYSIS: CHECKING DELAY STATUS (slightly DELAY)

```
print(df_small.loc[df_small['ARRIVAL_DELAY'].between(15,60), ['AIRLINE','ARRIVAL_DELAY']])
```

	AIRLINE	ARRIVAL_DELAY
27	NK	25.0
30	NK	43.0
35	HA	15.0
50	B6	20.0
73	US	60.0
...	...	...
5819011	B6	47.0
5819023	B6	20.0
5819051	AS	28.0
5819057	NK	28.0
5819071	B6	17.0

744347 rows

DATA ANALYSIS: CHECKING DELAY STATUS (medium DELAY)

```
filtered_df = df_small.loc[df_small['ARRIVAL_DELAY'].between(60, 120), ['AIRLINE', 'ARRIVAL_DELAY']]  
print(filtered_df)
```

	AIRLINE	ARRIVAL_DELAY
52	B6	85.0
55	B6	89.0
70	AA	102.0
73	US	60.0
86	AA	66.0
...	...	...
5818775	NK	104.0
5818814	EV	86.0
5818876	F9	119.0
5818886	UA	68.0
5818975	NK	100.0

[211171 rows x 2 columns]

211171rows

DATA ANALYSIS: CHECKING DELAY STATUS (HIGH DELAY)

```
print(df_small.loc[df_small['ARRIVAL_DELAY'].between(120,300), ['AIRLINE','ARRIVAL_DELAY']])
```

	AIRLINE	ARRIVAL_DELAY
106	DL	128.0
165	EV	226.0
174	UA	123.0
369	OO	144.0
493	MQ	161.0
...	...	...
5818593	EV	120.0
5818809	AA	228.0
5818836	DL	204.0
5819008	AA	234.0
5819069	B6	159.0

105214 rows

Data visualization

```
import matplotlib.pyplot as plt
import pandas as pd

dep_delay = df_small.groupby('AIRLINE')['DEPARTURE_DELAY'].mean()
arr_delay = df_small.groupby('AIRLINE')['ARRIVAL_DELAY'].mean()

airline_delay = pd.DataFrame({
    'Departure Delay': dep_delay,
    'Arrival Delay': arr_delay
}).sort_values(by='Arrival Delay')

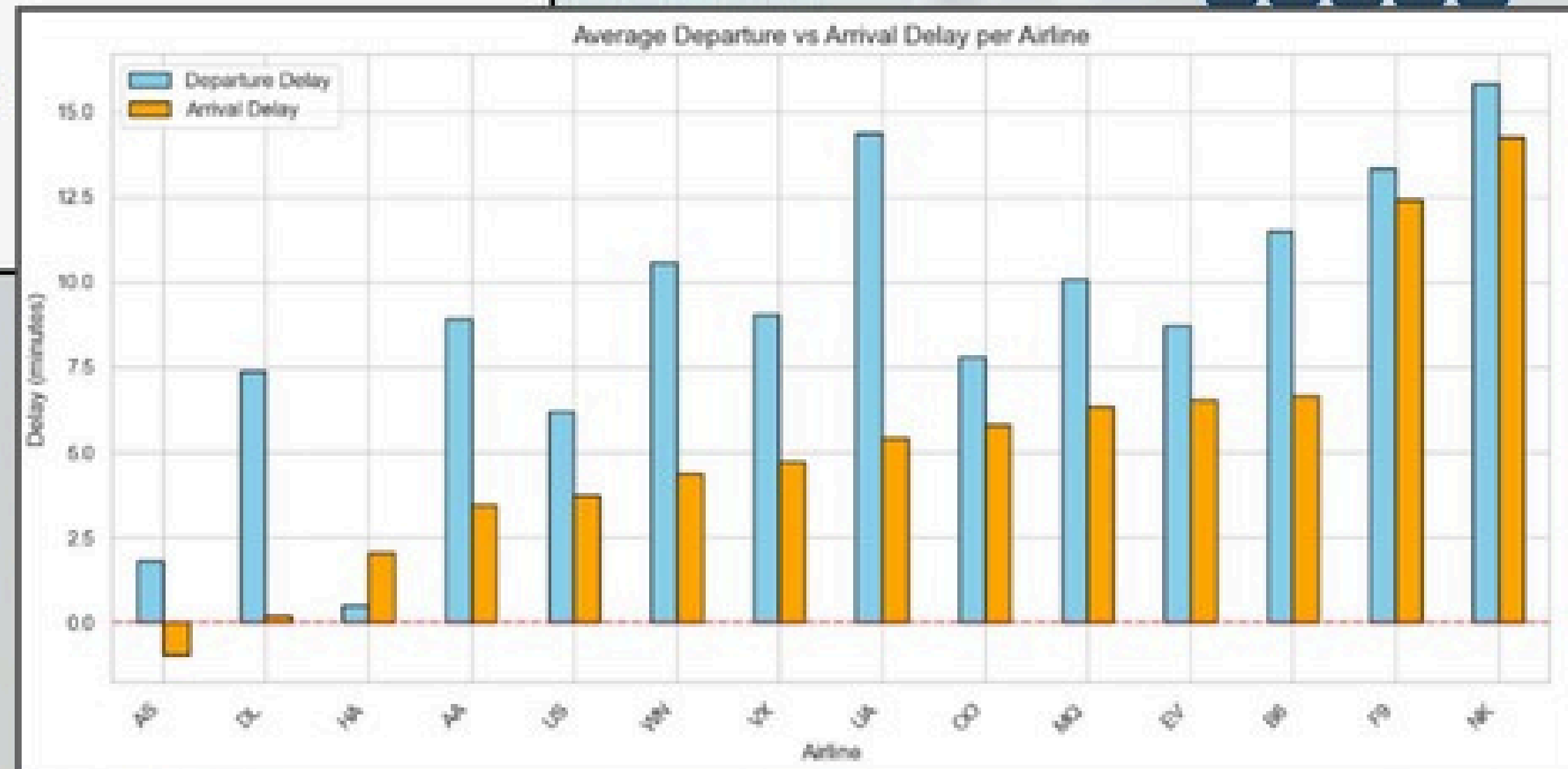
airline_delay.plot(
    kind='bar',
    figsize=(12, 4),
    color=['skyblue', 'orange'],
    edgecolor='black'
)

plt.title("Average Departure vs Arrival Delay per Airline", fontsize=14)
plt.xlabel("Airline", fontsize=12)
plt.ylabel("Delay (minutes)", fontsize=12)

plt.xticks(rotation=45, ha='right')
plt.axhline(0, color='red', linestyle='--', alpha=0.4)

plt.tight_layout()
plt.show()
```

NK and F9 recorded the highest average delays, whereas AS and HA performed best with the lowest levels. Interestingly, AS shows a negative arrival delay, meaning they often arrive earlier than scheduled.



Data visualization

```
import matplotlib.pyplot as plt

day_names = {
    1: 'Mon', 2: 'Tue', 3: 'Wed',
    4: 'Thu', 5: 'Fri', 6: 'Sat', 7: 'Sun'
}

df_small['DAY_NAME'] = df_small['DAY_OF_WEEK'].map(day_names)

arr_delay = df_small.groupby('DAY_NAME')['ARRIVAL_DELAY'].mean()
dep_delay = df_small.groupby('DAY_NAME')['DEPARTURE_DELAY'].mean()

day_delay = pd.DataFrame({
    'Arrival Delay': arr_delay,
    'Departure Delay': dep_delay
}).reindex(['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun'])

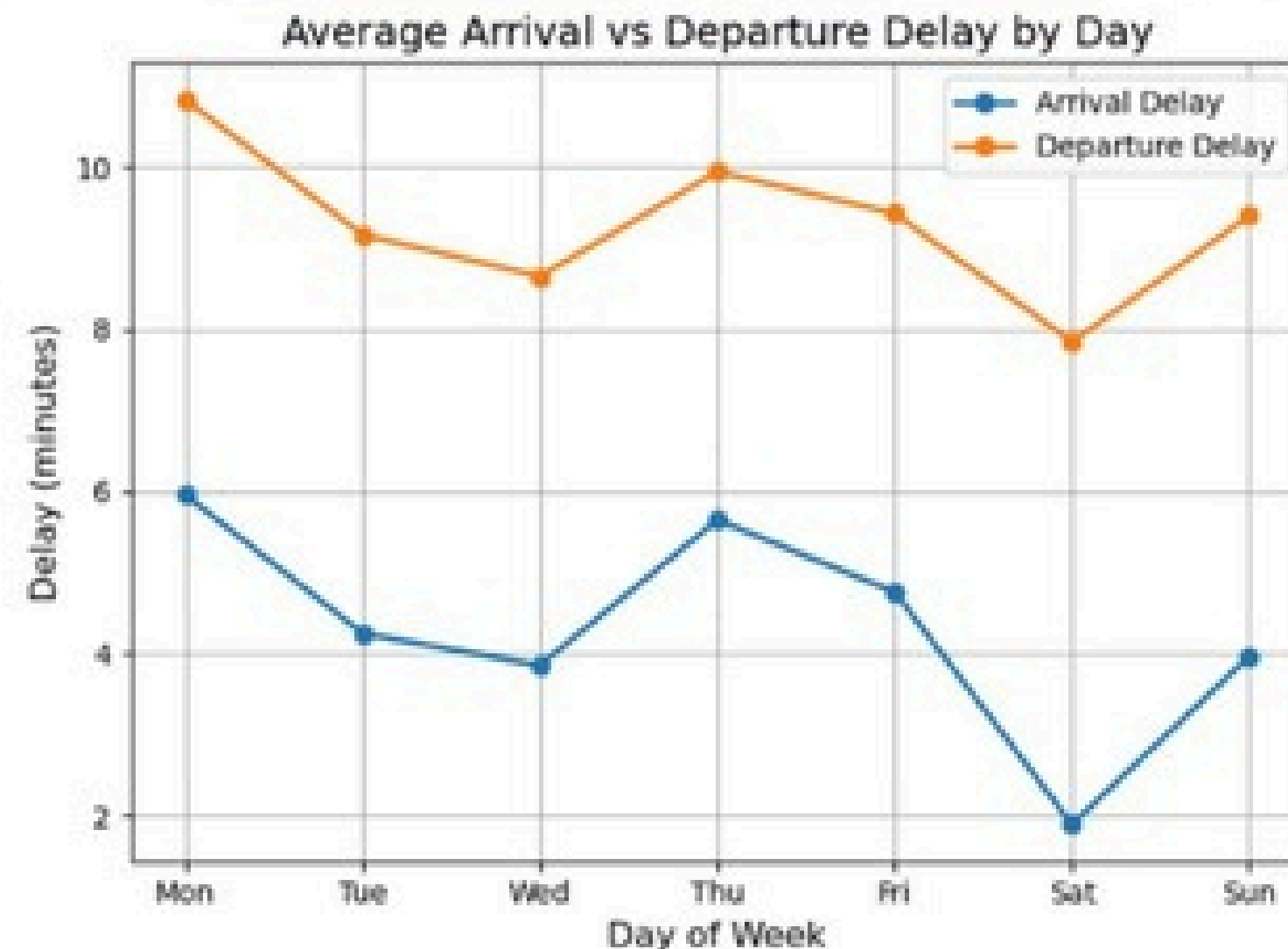
day_delay.plot(
    kind='line',
    marker='o',
    linewidth=2
)

plt.title("Average Arrival vs Departure Delay by Day", fontsize=14)
plt.xlabel("Day of Week", fontsize=12)
plt.ylabel("Delay (minutes)", fontsize=12)

plt.grid(True)
plt.tight_layout()
plt.show()
```

This line chart shows that delays vary across the week.

The highest average delay occurs on Monday, while the lowest is on Saturday.



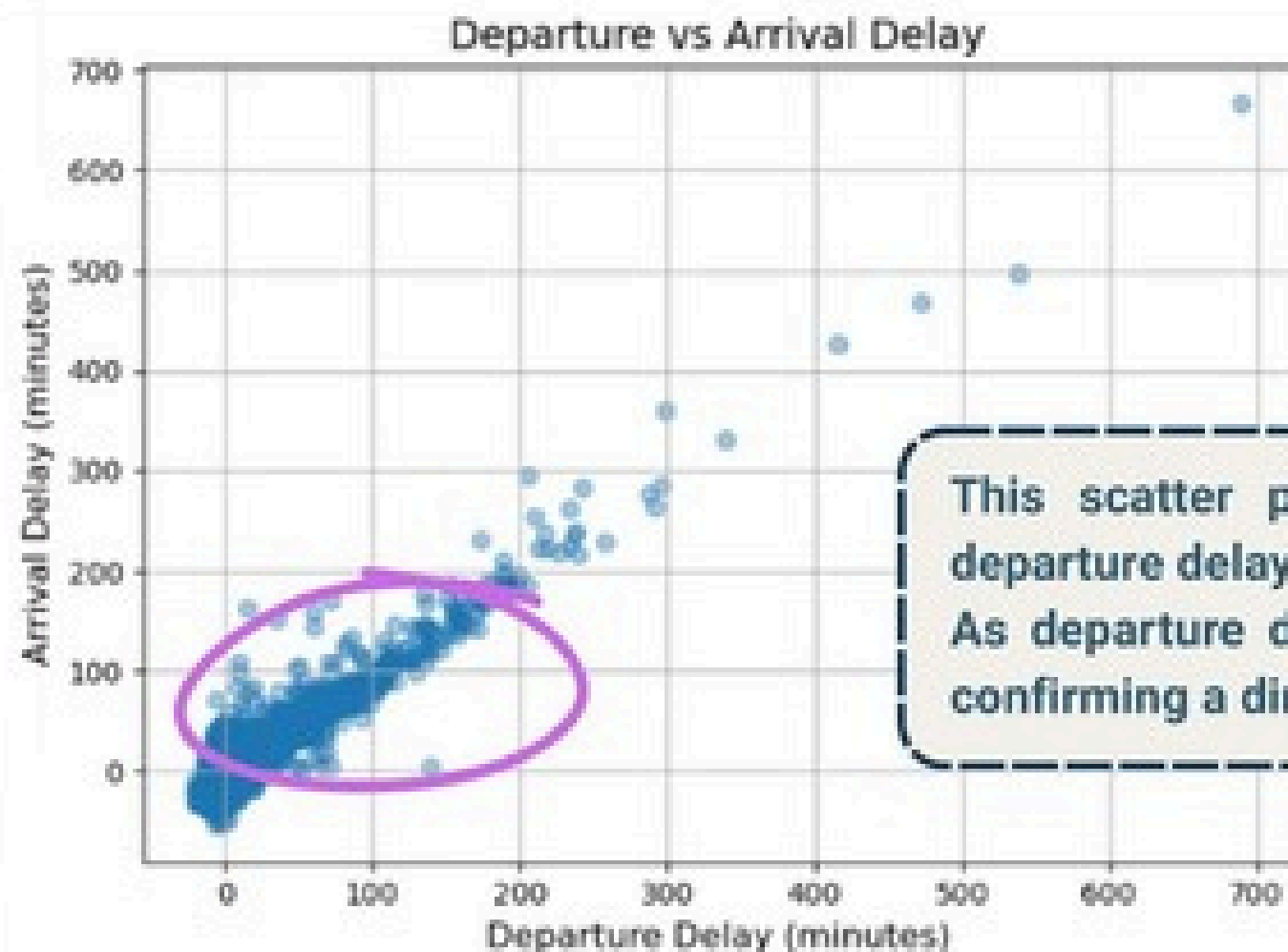
Data visualization

```
df_plot = df_small[['DEPARTURE_DELAY', 'ARRIVAL_DELAY']].dropna().sample(5000)

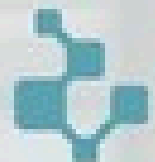
plt.scatter(
    df_plot['DEPARTURE_DELAY'],
    df_plot['ARRIVAL_DELAY'],
    alpha=0.4
)

plt.title("Departure vs Arrival Delay", fontsize=14)
plt.xlabel("Departure Delay (minutes)", fontsize=12)
plt.ylabel("Arrival Delay (minutes)", fontsize=12)

plt.grid(True)
plt.tight_layout()
plt.show()
```



This scatter plot shows a strong positive relationship between departure delay and arrival delay. As departure delay increases, arrival delay also tends to increase, confirming a direct correlation between both variables.



Data visualization



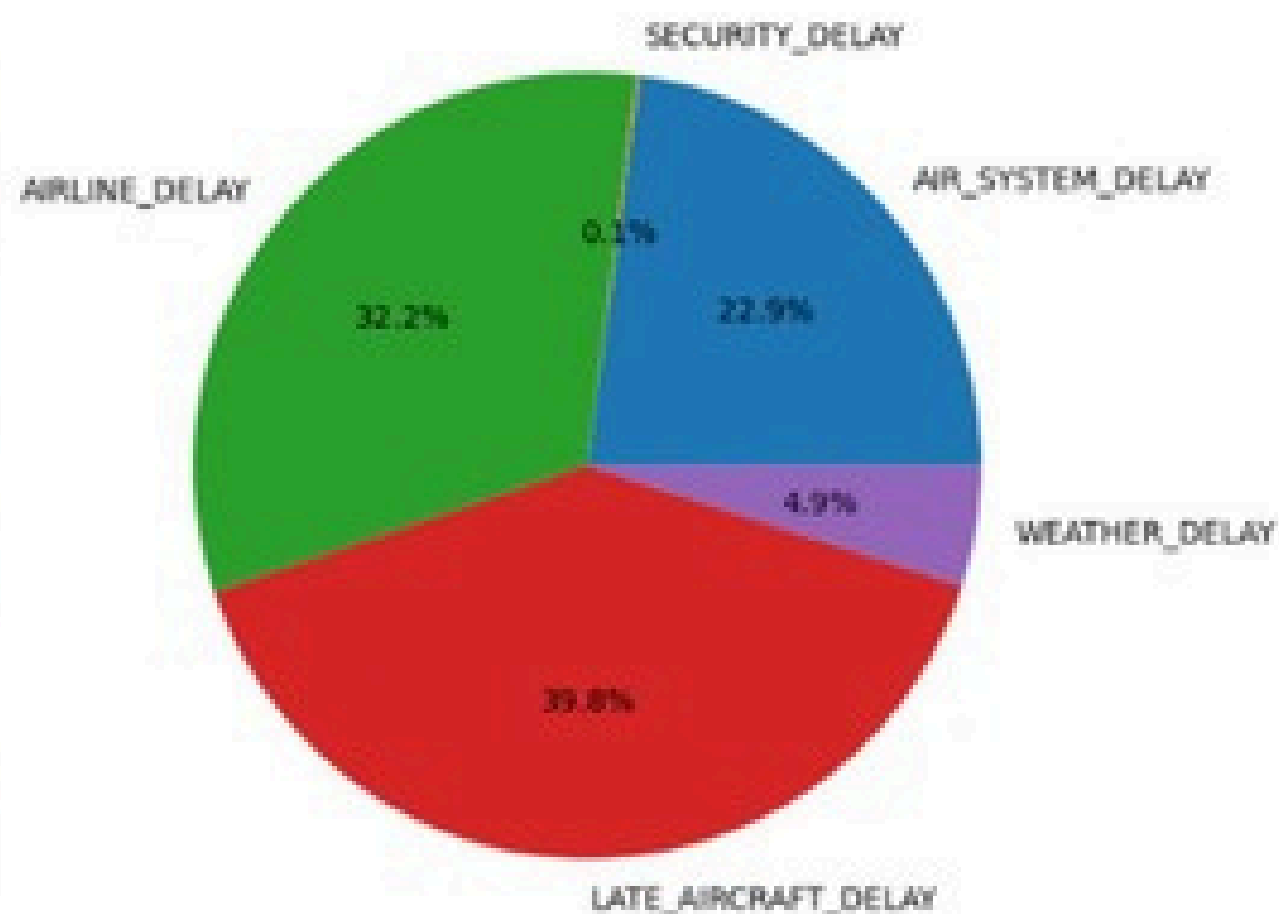
```
delay_causes = df[['AIR_SYSTEM_DELAY', 'SECURITY_DELAY', 'AIRLINE_DELAY', 'LATE_AIRCRAFT_DELAY', 'WEATHER_DELAY']].sum()

plt.figure(figsize=(6,6))

delay_causes.plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title("Distribution of Delay Causes", fontsize=14)
plt.ylabel("")
plt.tight_layout()
plt.show()
```

Distribution of Delay Causes



The pie chart shows that the main cause of delays is late aircraft, contributing the largest percentage. The least contributing factor is weather delay. This highlights that operational and system issues have more impact than external factors like security.



Data visualization

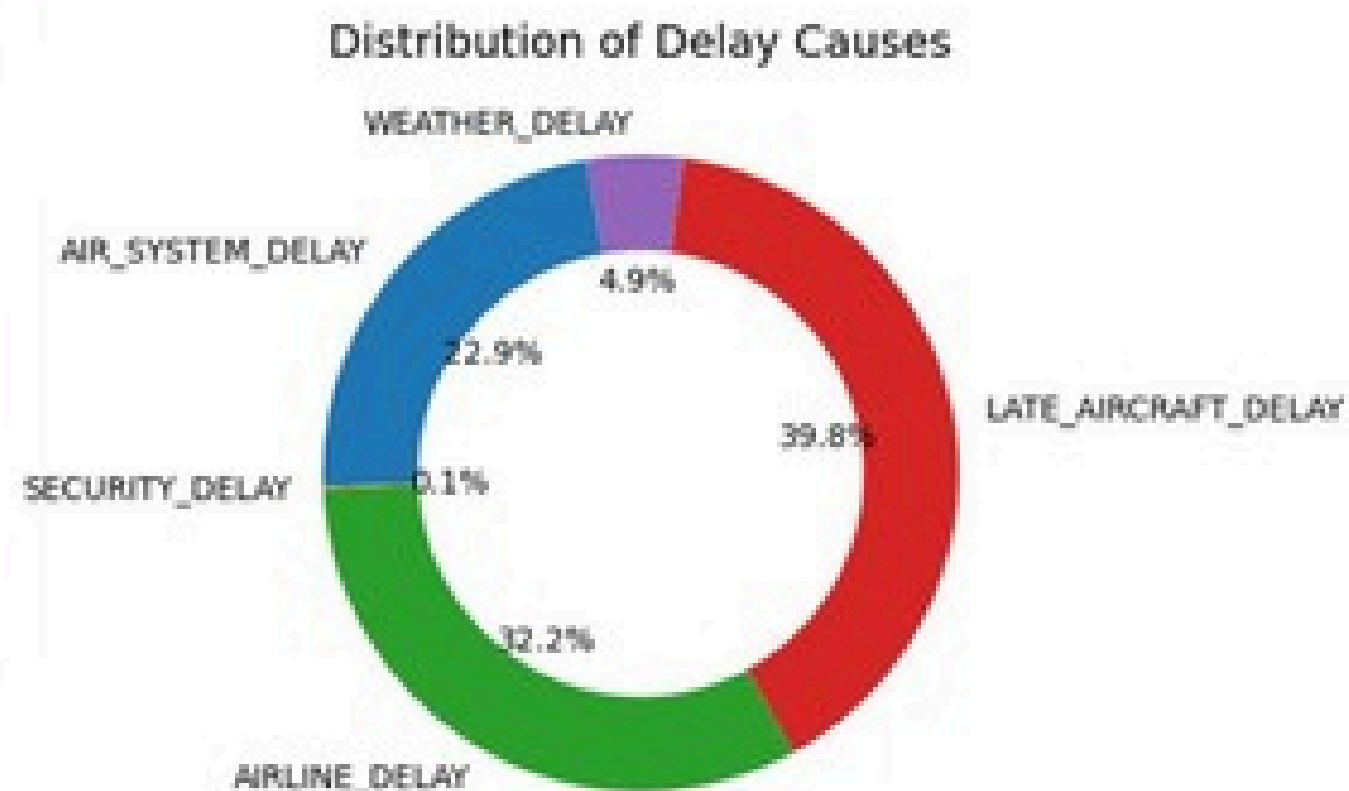


```
delay_causes = df[['AIR_SYSTEM_DELAY', 'SECURITY_DELAY', 'AIRLINE_DELAY', 'LATE_AIRCRAFT_DELAY', 'WEATHER_DELAY']].sum()

plt.figure(figsize=(6,6))

delay_causes.plot(
    kind='pie',
    autopct='%1.1f%%',
    startangle=100,
    wedgeprops={'width': 0.3}
)

plt.title("Distribution of Delay Causes", fontsize=14)
plt.ylabel("")
plt.tight_layout()
plt.show()
```



Late Aircraft Delay and Air System Delay are the dominant factors, accounting for the largest portions of the chart.





Data visualization



```
In [51]: df_hist = df_small['ARRIVAL_DELAY'].dropna().sample(5000)
```

```
plt.hist(df_hist, bins=30)
```

```
plt.title("Distribution of Arrival Delay", fontsize=14)
```

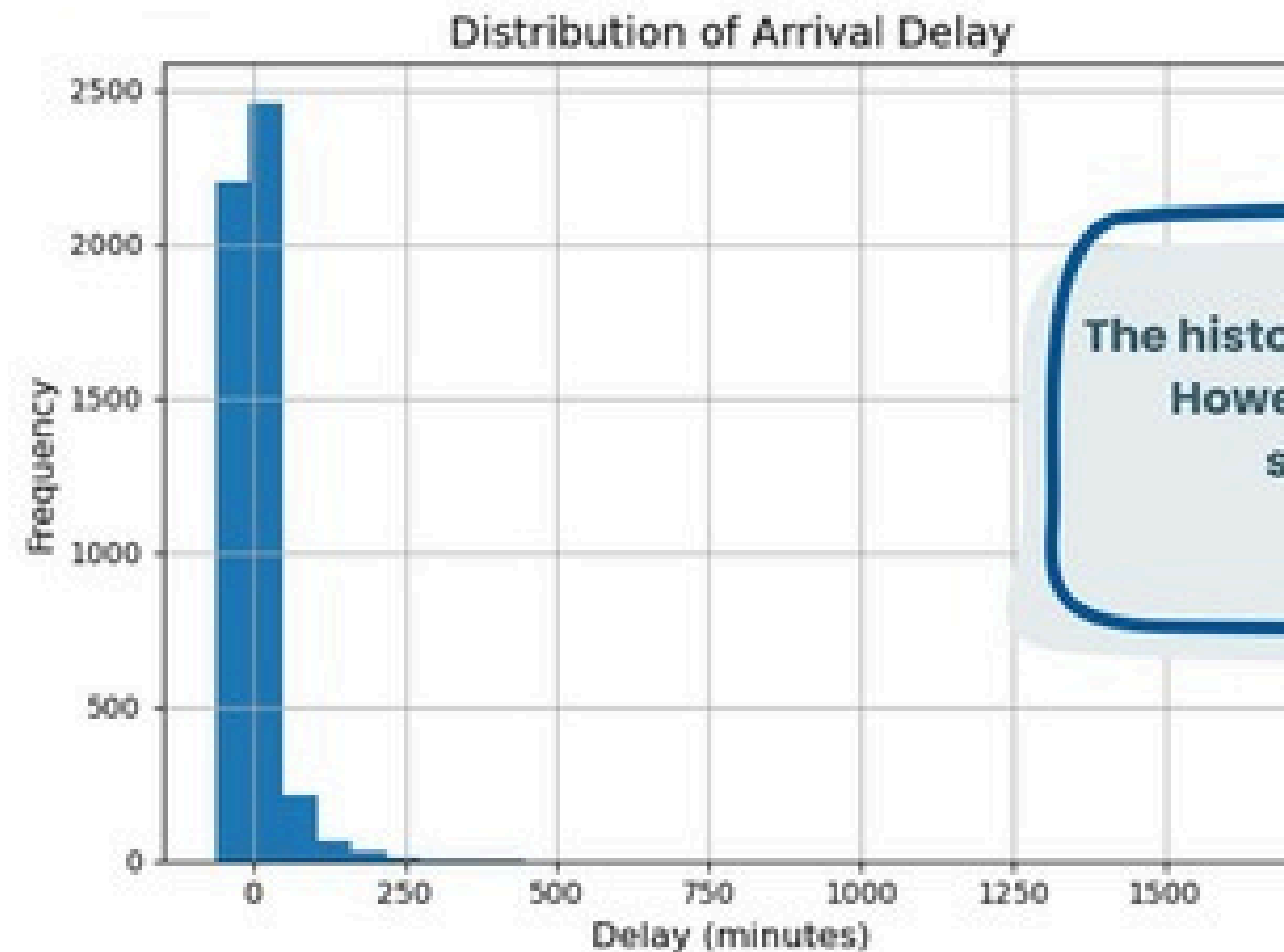
```
plt.xlabel("Delay (minutes)", fontsize=12)
```

```
plt.ylabel("Frequency", fontsize=12)
```

```
plt.grid(True)
```

```
plt.tight_layout()
```

```
plt.show()
```



The histogram shows that most flights have small delays close to 0. However, there are some extreme values indicating rare but significant delays, which affect the overall average.



Data visualization



```
import matplotlib.pyplot as plt

df_box = df_small['ARRIVAL_DELAY'].dropna().sample(5000)
q1 = df_box.quantile(0.25)
q2 = df_box.quantile(0.50)
q3 = df_box.quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
print("Q1:", q1)
print("Q2 (Median):", q2)
print("Q3:", q3)
print("Lower bound:", lower)
print("Upper bound:", upper)

plt.figure(figsize=(6,5))
plt.boxplot(df_box)

plt.title("Box Plot of Arrival Delay", fontsize=14)
plt.ylabel("Delay (minutes)", fontsize=12)

plt.grid(True)
plt.tight_layout()
plt.show()
```

```
Q1: -13.0
Q2 (Median): -4.0
Q3: 7.0
Lower bound: -43.0
Upper bound: 37.0
```

- **Median (-4.0):**
Indicates that 50% of the flights actually arrive 4 minutes early, showing good overall performance.
- **IQR (Q1 to Q3):** The core 50% of flights fall between a 13-minute early arrival and a 7-minute delay.
- **Outlier Thresholds:** The Upper Bound is 37.0 minutes, The lower Bound is -43 minutes.



Data visualization

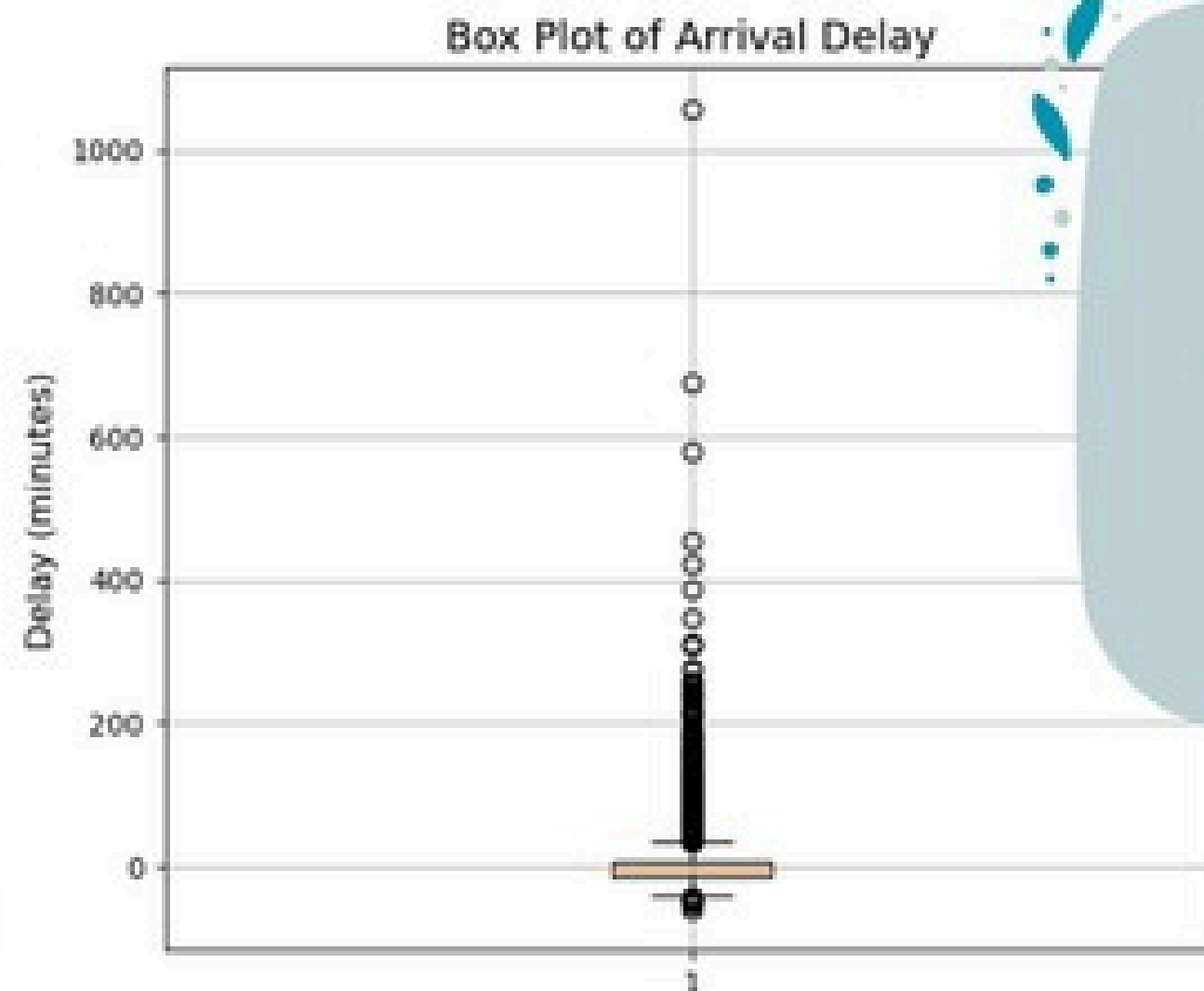
```
] df_box = df_small['ARRIVAL_DELAY'].dropna().sample(5000)

plt.figure(figsize=(6,5))

plt.boxplot(df_box)

plt.title("Box Plot of Arrival Delay", fontsize=14)
plt.ylabel("Delay (minutes)", fontsize=12)

plt.grid(True)
plt.tight_layout()
plt.show()
```



- The box plot shows the distribution of arrival delays and highlights several outliers.
- These outliers represent flights with unusually high delays, which significantly affect the dataset.

Data visualization

```
import matplotlib.pyplot as plt

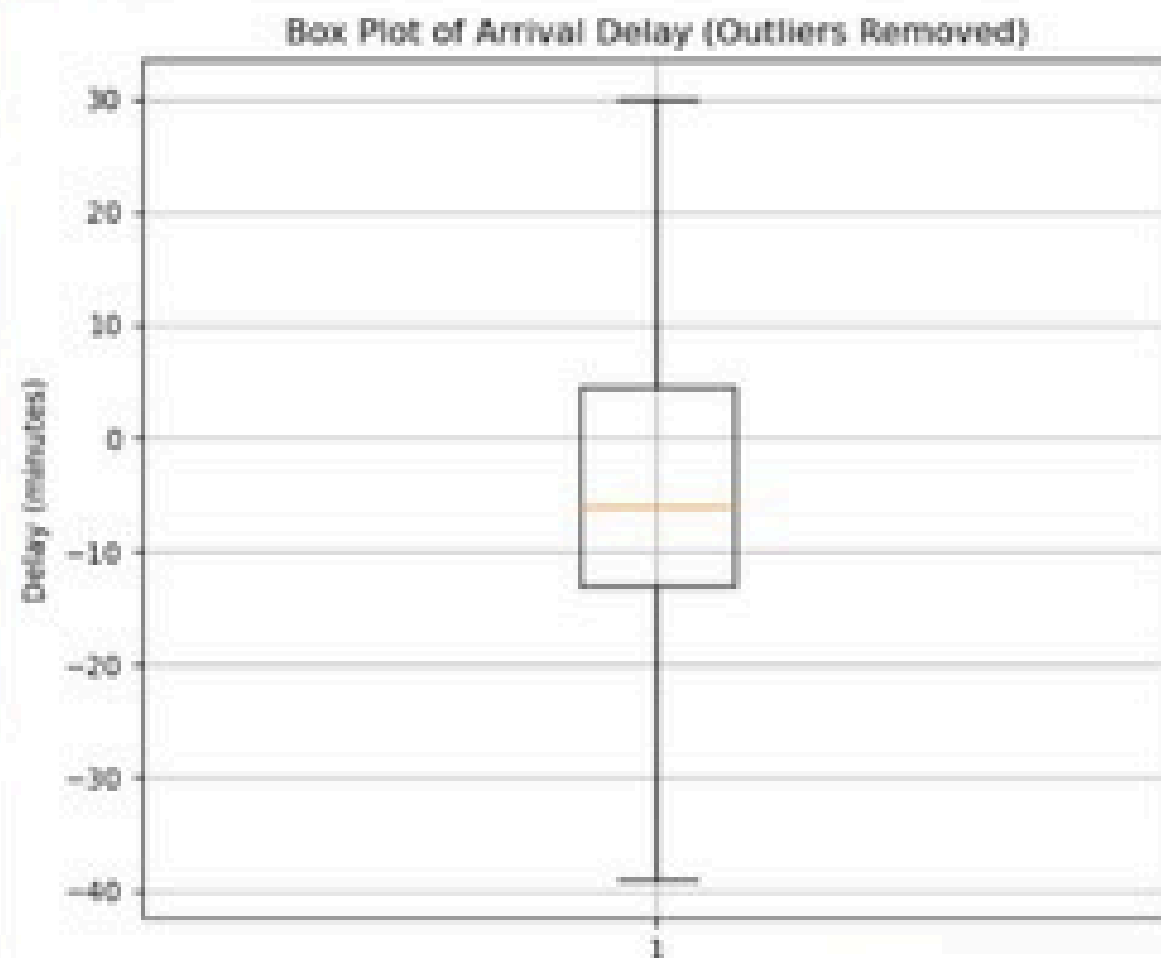
df_box = df_small['ARRIVAL_DELAY'].dropna().sample(5000, random_state=42)

q1 = df_box.quantile(0.25)
q3 = df_box.quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr

df_clean = df_box[(df_box >= lower) & (df_box <= upper)]

plt.figure(figsize=(6,5))
plt.boxplot(df_clean, showfliers=False)

plt.title("Box Plot of Arrival Delay (Outliers Removed)")
plt.ylabel("Delay (minutes)")
plt.grid(True)
plt.tight_layout()
plt.show()
```



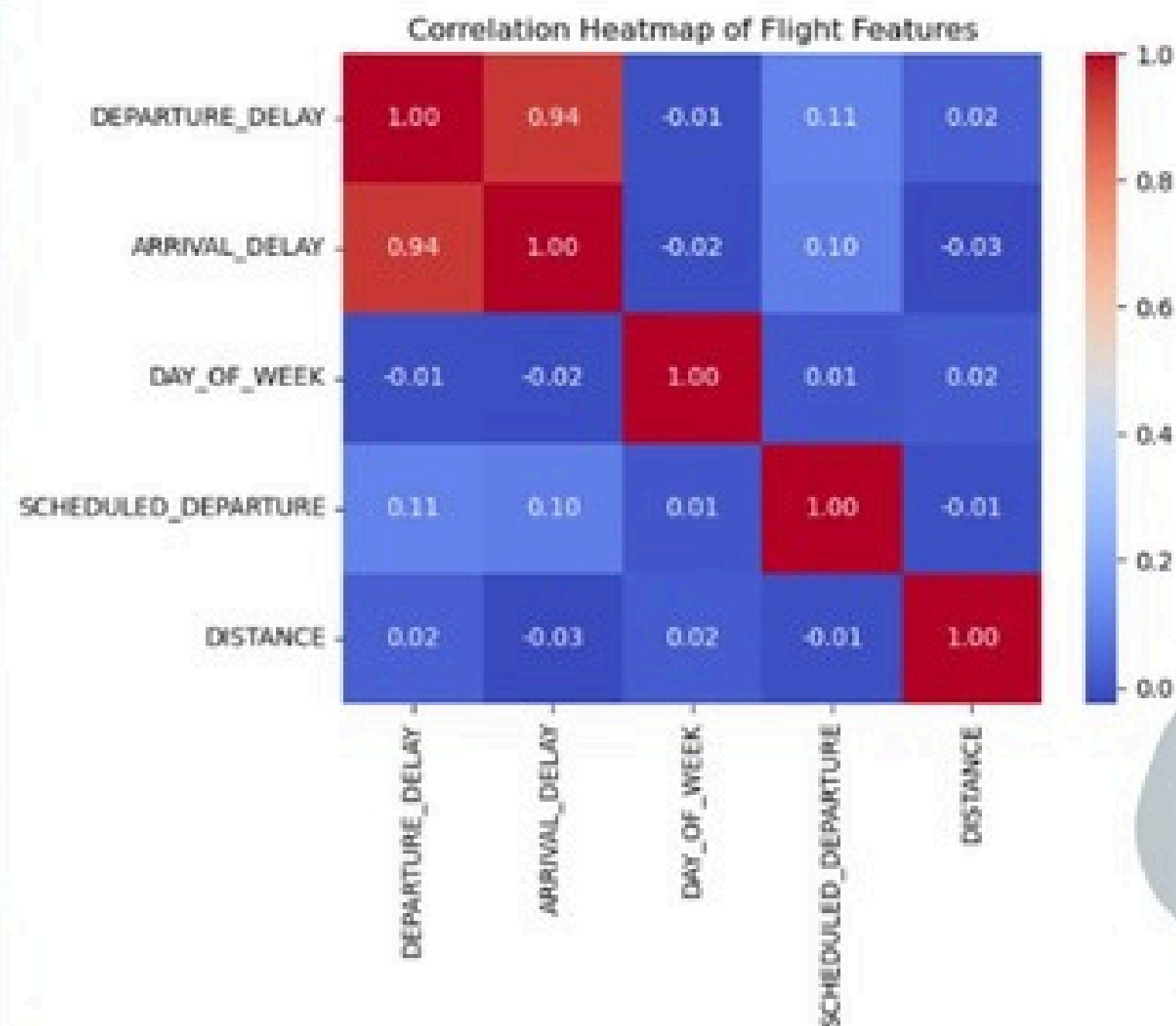
The wider upper section of the box indicates more variability in delay times, while the narrower lower section shows that early arrivals are more consistent and stable.



Data visualization



```
import seaborn as sns
import matplotlib.pyplot as plt
corr = df_small.corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap of Flight Features')
plt.show()
```



There is a very strong positive correlation (0.94) between DEPARTURE_DELAY and ARRIVAL_DELAY.

Data visualization

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df_small = df_small.copy()
df_small['hour'] = (df_small['SCHEDULED_DEPARTURE'] // 100).astype(int)

day_map = {
    1: "Mon", 2: "Tue", 3: "Wed",
    4: "Thu", 5: "Fri", 6: "Sat", 7: "Sun"
}
df_small['day_of_week'] = df_small['hour'].map(day_map)

order = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
df_small['day_of_week'] = pd.Categorical(
    df_small['day_of_week'],
    categories=order,
    ordered=True
)

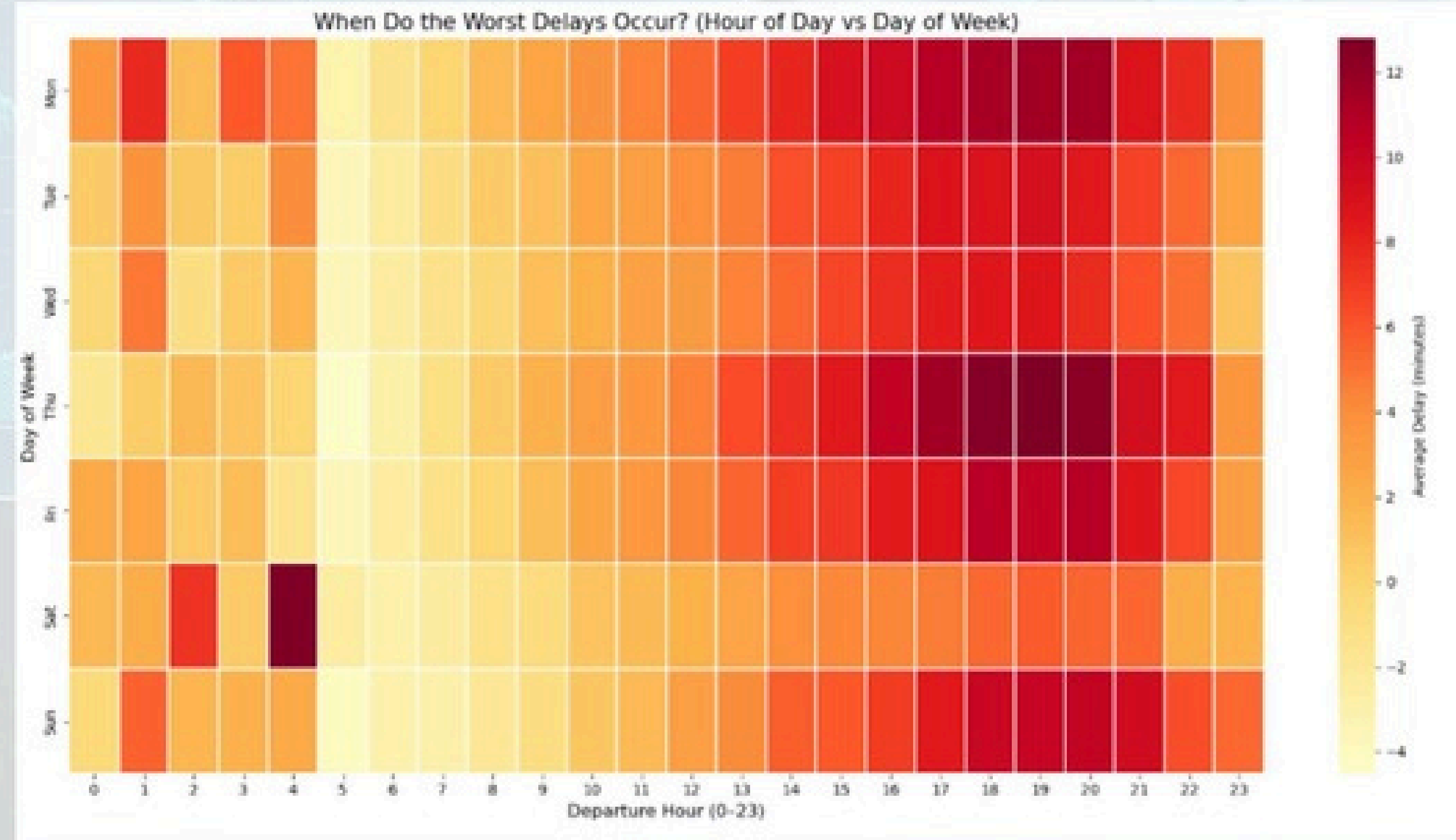
pivot_table = df_small.pivot_table(
    index='day_of_week',
    columns='hour',
    values='ARRIVAL_DELAY',
    aggfunc='mean'
)

plt.figure(figsize=(15, 8))
sns.heatmap(
    pivot_table,
    cmap="magma",
    annot=False,
    linewidths=0.5,
    linecolor='white',
    cbar_kws={'label': 'Average delay (minutes)'}
)

plt.title('When Do the Worst Delays Occur? (Hour of Day vs Day of Week)', fontsize=12)
plt.xlabel('Departure Hour (0-23)', fontsize=12)
plt.ylabel('Day of Week', fontsize=12)

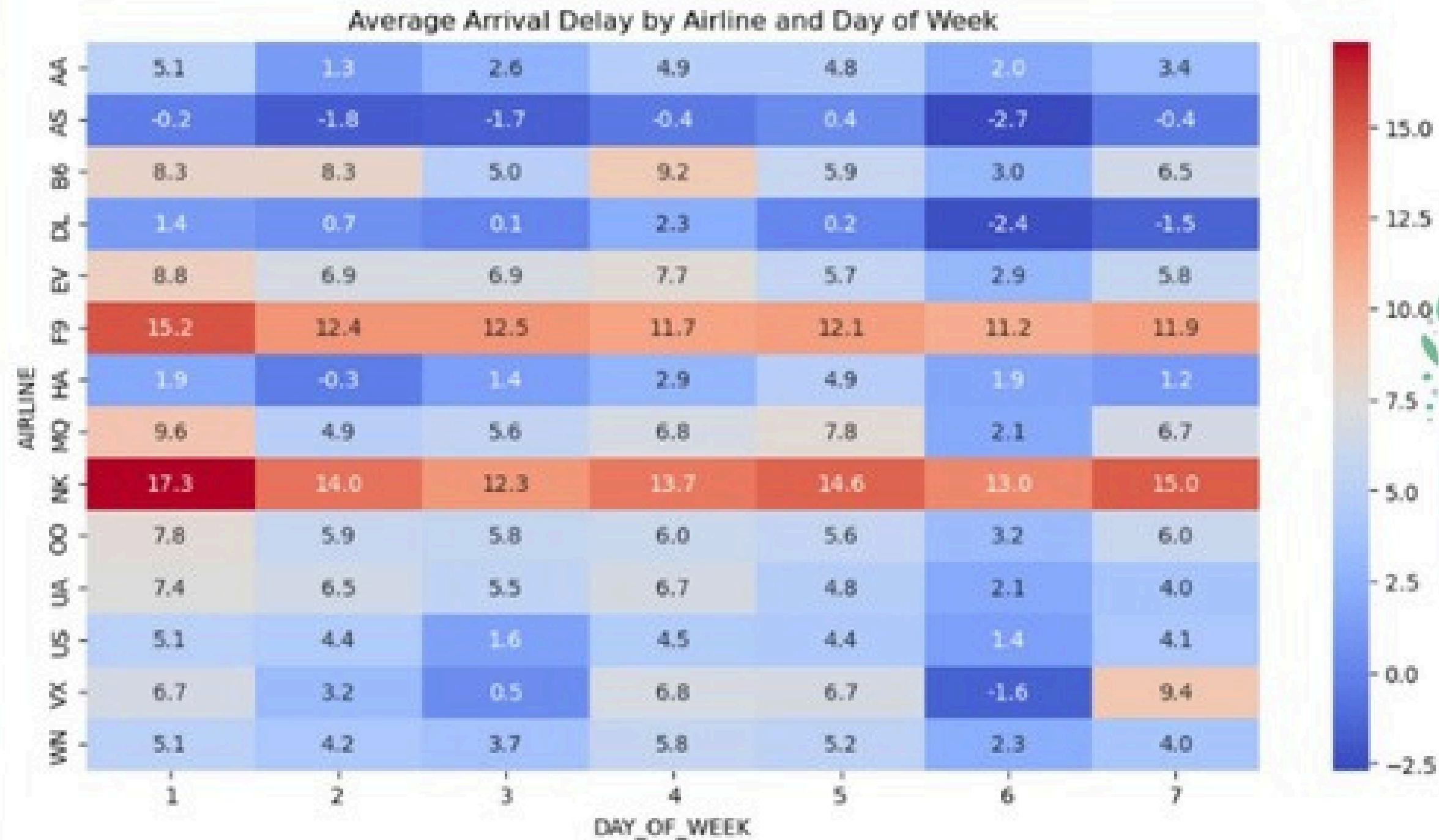
plt.tight_layout()
plt.show()
```

This heatmap highlights the peak periods of flight delays throughout the day and week. It shows that delays are most concentrated during afternoon and evening hours.



Data visualization

```
import seaborn as sns
import matplotlib.pyplot as plt
result = df_small.groupby(['AIRLINE', 'DAY_OF_WEEK'])['ARRIVAL_DELAY'].mean().unstack()
sns.heatmap(result, annot=True, fmt=".1f", cmap="coolwarm")
plt.title("Average Arrival Delay by Airline and Day of Week")
plt.show()
```



This heatmap shows the average flight delays for each airline across different days. **Red** means higher delays, while **blue** means lower or early arrivals.

Data visualization

```
import plotly.express as px

df_sample = df_small.sample(50, random_state=42).copy()

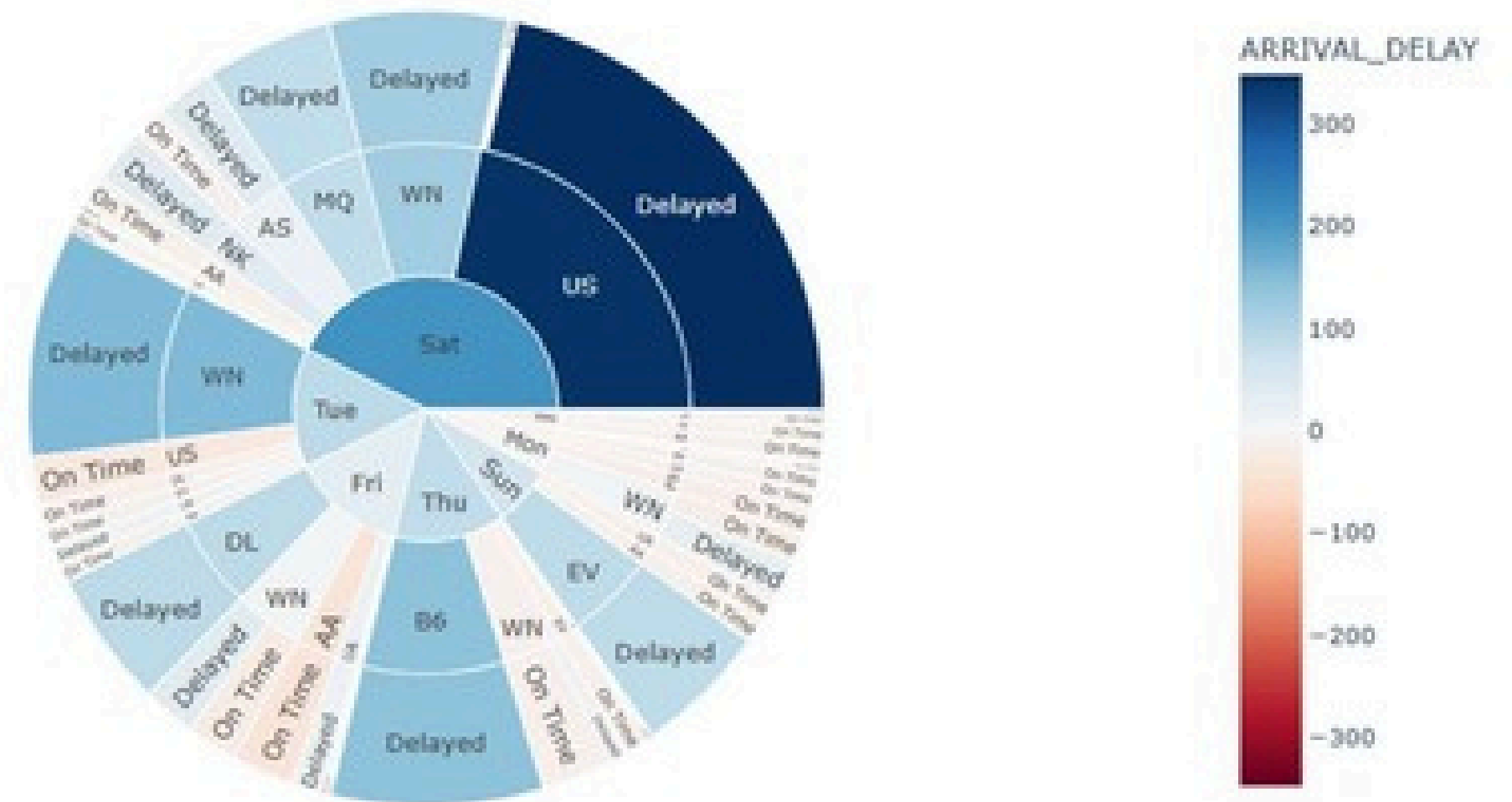
#
df_sample = df_sample.dropna(subset=[
    'DAY_OF_WEEK', 'AIRLINE', 'DELAY_STATUS', 'ARRIVAL_DELAY'
])

df_sample['DELAY_SIZE'] = df_sample['ARRIVAL_DELAY'].abs()
df_sample = df_sample[df_sample['DELAY_SIZE'] > 0]
day_map = {1:"Mon",2:"Tue",3:"Wed",4:"Thu",5:"Fri",6:"Sat",7:"Sun"}
df_sample['DAY_OF_WEEK'] = df_sample['DAY_OF_WEEK'].map(day_map)
fig = px.sunburst(
    df_sample,
    path=['DAY_OF_WEEK', 'AIRLINE', 'DELAY_STATUS'],
    values='DELAY_SIZE',
    color='ARRIVAL_DELAY',
    color_continuous_scale='RdBu',
    color_continuous_midpoint=0
)

fig.update_layout(title="Hierarchical Analysis of Flight Delays")
fig.show()
```

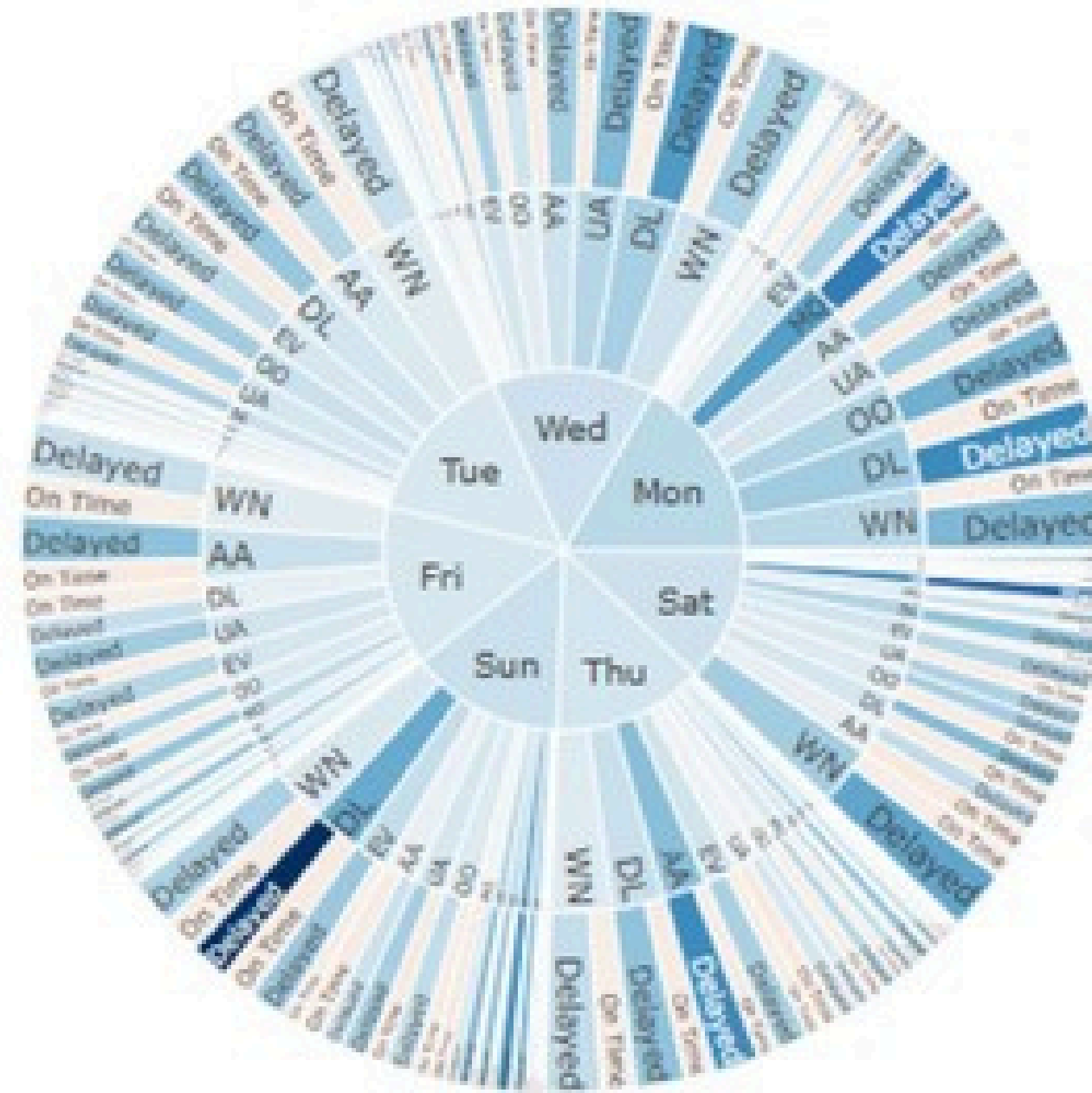
This hierarchical chart shows flight delays distributed by day of the week, airline, and flight status. It helps identify which days and airlines have higher delayed flights compared to on-time flights.

Hierarchical Analysis of Flight Delays



Data visualization

Hierarchical Analysis of Flight Delays



ARRIVAL_DELAY

200

100

0

-100

-200

we use 7000 samples

Data visualization

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# copy to avoid SettingWithCopyWarning
df_small = df_small.copy()

day_names = {
    0: 'Mon', 1: 'Tue', 2: 'Wed',
    3: 'Thu', 4: 'Fri', 5: 'Sat', 6: 'Sun'
}
df_small['DAY_NAME'] = df_small['DAY_OF_WEEK'].map(day_names)

df_small = df_small.dropna(subset=['DAY_NAME', 'ARRIVAL_DELAY'])

df_small = df_small[
    (df_small['ARRIVAL_DELAY'] > -50) &
    (df_small['ARRIVAL_DELAY'] < 50)
]
(style='whitegrid')

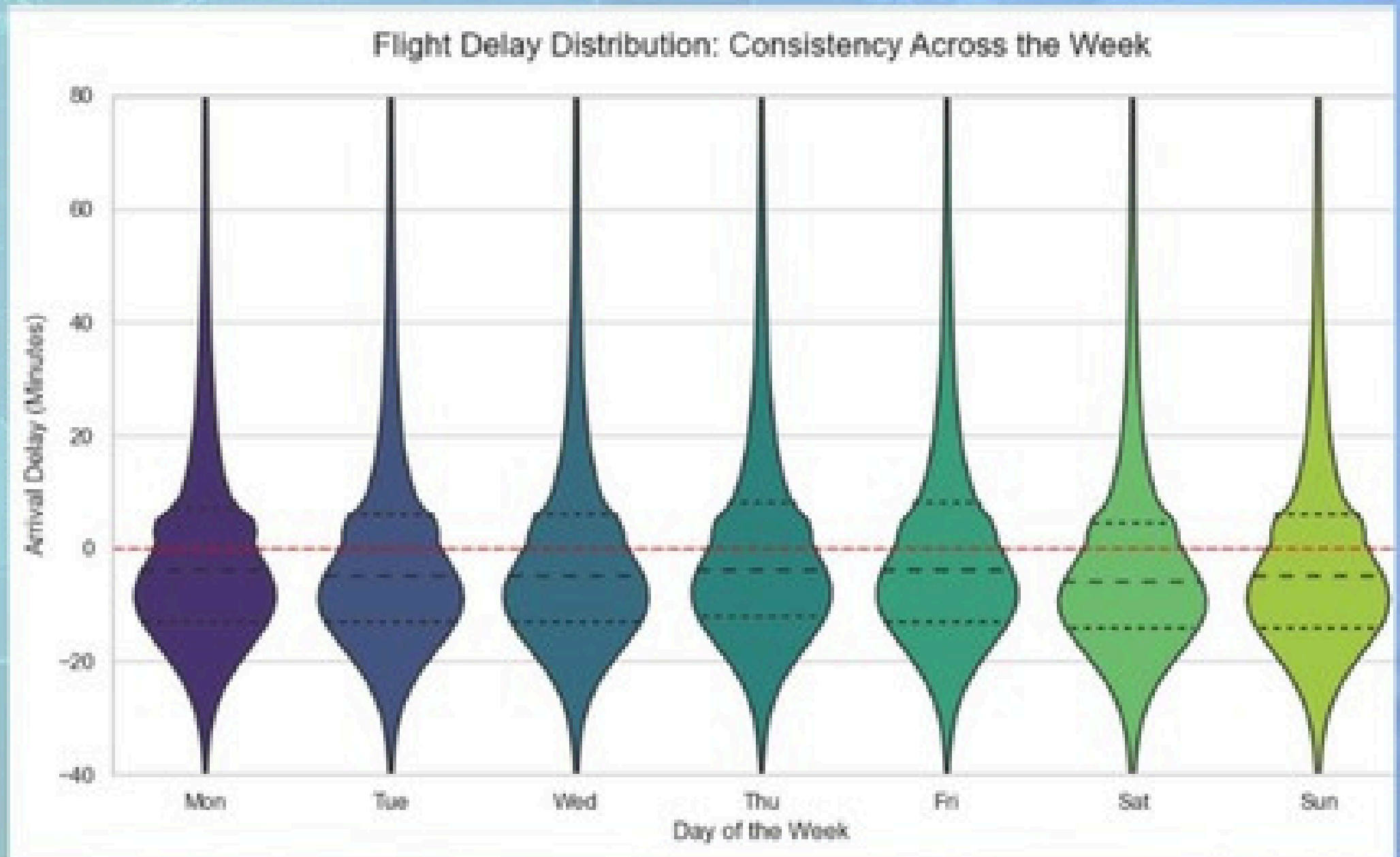
plot_order = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']

plt.figure(figsize=(18, 6))
ax = sns.violinplot(
    x='DAY_NAME',
    y='ARRIVAL_DELAY',
    data=df_small,
    order=plot_order,
    palette='viridis',
    inner='quartile',
    cut=0
)

plt.ylim(-40, 80)

plt.title('Flight Delay Distribution: Consistency Across the Week', fontsize=14, pad=20)
plt.xlabel('Day of the week', fontsize=12)
plt.ylabel('Arrival Delay (Minutes)', fontsize=12)
plt.axline(0, color='red', linestyle='--', alpha=0.4)

plt.tight_layout()
plt.show()
```



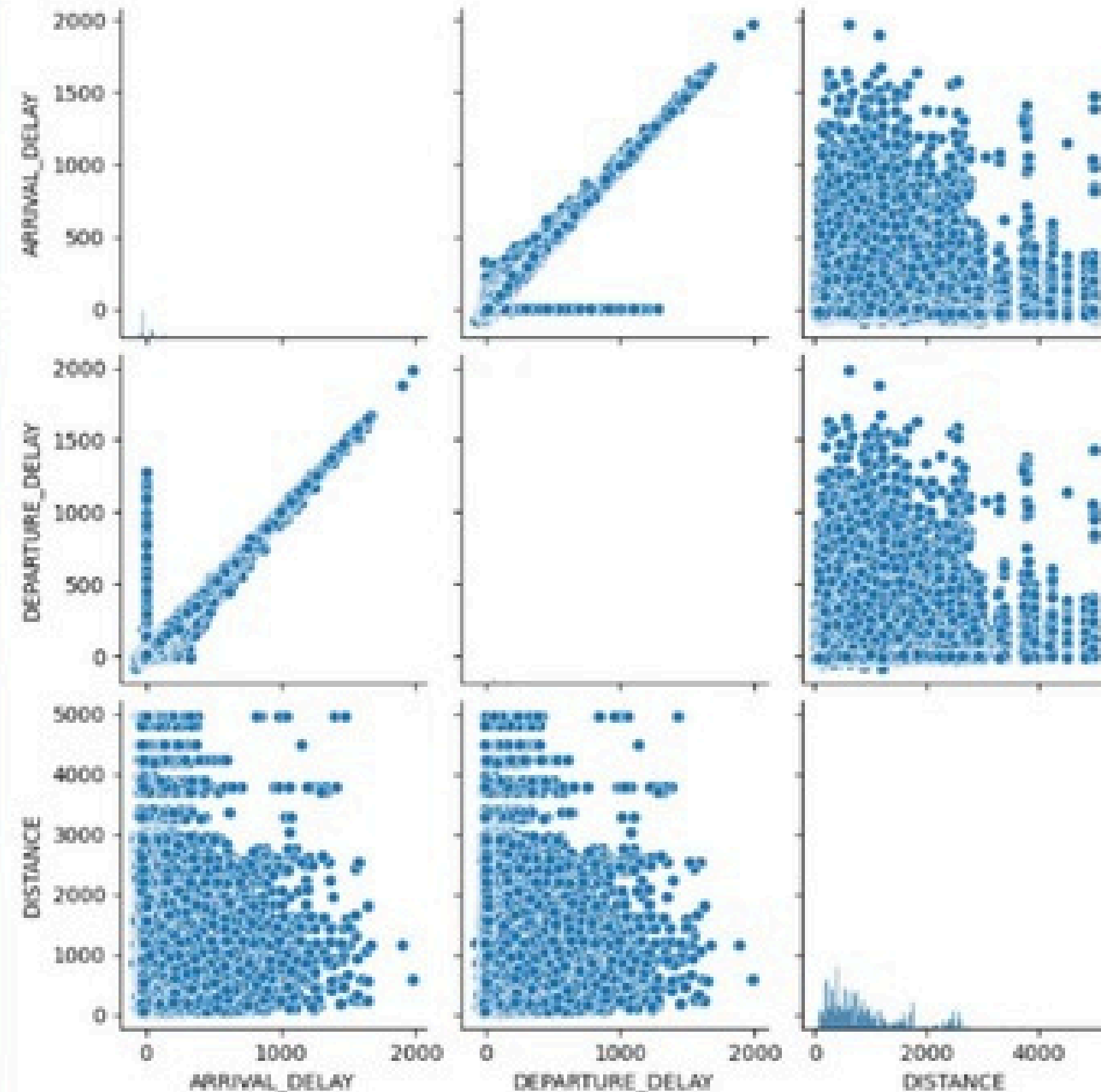
- Flight delays are generally consistent across all days of the week, with no major differences in average delay patterns.
- Weekend flights (Saturday and Sunday) show slightly wider distributions, indicating greater variability and a higher chance of irregular delays.
- The long upper tails in the plot suggest that extreme delays can occur on any day, although such cases are relatively uncommon.
- Overall, the variation in delay spread is more noticeable than the difference in average delay between weekdays.

Data visualization

- **Strong positive correlation:**
- Departure delay directly increases arrival delay (clear linear pattern).
- **Most delays are small:**
- Majority of flights have delays under ~500 minutes; large delays are rare.
- **Distance $\neq$ delay:**
- No clear relationship between flight distance and delays.



```
sns.pairplot(df_small[['ARRIVAL_DELAY', 'DEPARTURE_DELAY', 'DISTANCE']])  
plt.show()
```



Final Insights:

- The **Root Cause**: A powerful correlation of **0.94** exists between Departure and Arrival delays. This proves that punctuality starts on the ground, not in the air.
- After cleaning outliers, we found a **Negative Median** . This means **more than 50%** of flights actually arrive **earlier** than scheduled.
- **Late Aircraft** and National Air System are the primary delay factors.
- **Monday** was identified as the most challenging day, showing the highest average delays and the widest spread of inconsistency.
- **Alaska Airlines (AS)** stands out as the most punctual carrier, while **Spirit Airlines (NK)** faces the highest frequency of significant delays.
- **To improve** performance, airlines should focus on Monday staffing and increasing "buffer times" between connecting flights.



Recommendations:

For Airports:

- Upgrade air traffic control systems to manage bad weather.
- Expand runway capacity to reduce waiting queues.

For Future Research:

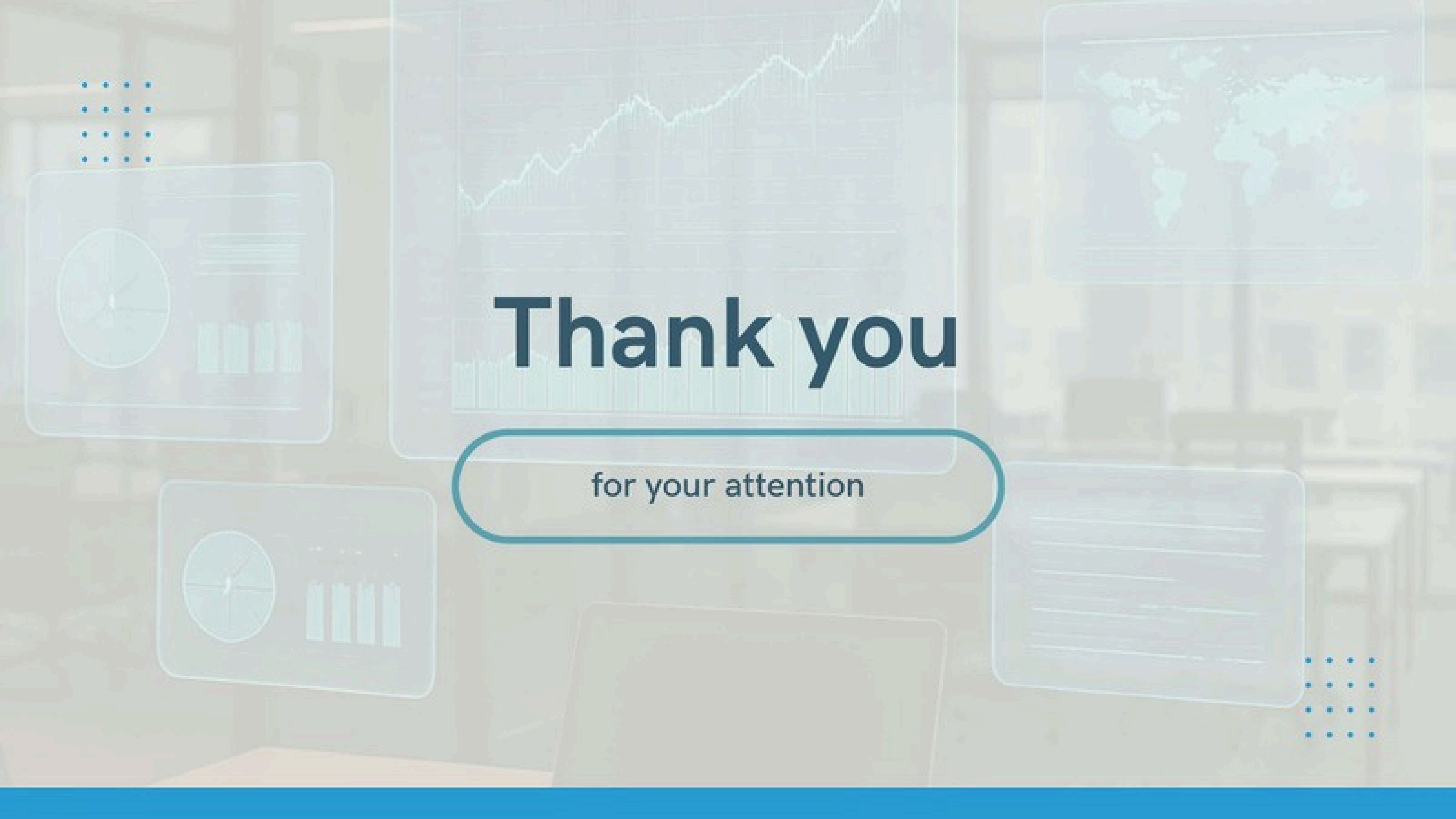
- Study how specific weather types affect delay duration.
- Analyze how holiday seasons impact airline performance.

For Airlines:

- Optimize flight schedules on busy days like Mondays.
- Increase buffer time between flights to prevent delay spread.

For Passengers:

- Book flights on Saturdays to avoid long delays.
- Choose airlines with stable schedules like AS or HA.

The background is a blurred office scene with people working at desks. Overlaid on this are several semi-transparent blue boxes containing various data visualizations: a line graph showing an upward trend, a world map, a pie chart, and bar charts. In the top-left and bottom-right corners, there are decorative 4x4 grids of small blue dots.

Thank you

for your attention